CS-30003 · Coding Assignment 1 · Project Proposal

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Repository: https://github.com/fabricnp/bittorrent-lite

Project: P8 · P2P File Distribution (BitTorrent-lite)

Path: A (catalogue)

1. WHAT WE ARE BUILDING

We are building a BitTorrent-lite peer-to-peer file distribution system from the socket level. The system has three components: a tracker that maintains swarm membership and serves peer lists over a custom HTTP interface built on raw TCP sockets, a metainfo generator that splits files into fixed-size pieces and produces per-piece SHA-256 hashes, and a peer client that implements a fully specified wire protocol (handshake, bitfield, have, request, piece, cancel) over TCP sockets for concurrent multi-peer file transfer. It operates at the application layer, directly on top of TCP.

2. CORE DELIVERABLES

- ☐ Metainfo file format: configurable piece sizes with per-piece SHA-256 hashes in Bencode-like binary format
- ☐ Tracker server (HTTP over raw TCP sockets): peer registration, announce, swarm state, peer list responses
- ☐ Peer wire protocol: handshake, bitfield, have, request/piece/cancel — documented in a protocol spec
- ☐ Rarest-first piece selection with random fallback; request pipelining (up to 5 outstanding requests/peer); per-piece hash verification
- ☐ Multi-peer operation: ≥6 concurrent peers, 1–2 seeders, per-peer async state machines (Python asyncio)
- ☐ Basic choking/unchoking mechanism to manage peer upload opportunities and handle free-riding peers
- ☐ End-to-end integrity: SHA-256 of reassembled file matches original

3. THE CLAIM WE WILL TEST

We hypothesize that rarest-first piece selection reduces total swarm distribution time compared with random and sequential selection, particularly as swarm size increases. Experiment: fix a 100 MB file and 256 KB piece size; vary swarm size (6 and 8 peers) and seeder count (1 and 2), and compare the three piece-selection strategies under the same rate-limit conditions. IV: piece selection strategy, swarm size, and seeder count. DVs: total distribution time, per-peer completion time, and piece availability variance. 5 runs per configuration; results as box plots. Second experiment: 2 free-rider peers (never upload) to evaluate choking effectiveness.

4. STACK AND JUSTIFICATION

Python 3.12 — asyncio (non-blocking sockets), hashlib (SHA-256), struct (binary framing), matplotlib + pandas (plots). Python's asyncio provides clean peer state management without threads; rate-limited experiments make throughput ceiling irrelevant.

5. WORK SPLIT

Owner	Component
Member A	Metainfo format, file splitter/assembler, hashing, file reassembly
Member B	Tracker server (raw TCP HTTP, announce, swarm state)
Member C	Peer wire protocol (message framing, handshake, state machine)
Member D	Piece selection algorithms, choking logic, experiment harness, analysis
A + C	Integration, multi-peer orchestration, E2E testing
B + D	Protocol specification document

6. RISKS

1. Concurrency bugs under 6+ peers — deadlocks in async state machines. Fallback: start 3-peer tests from week 3; structured logging; drop to thread-per-peer if asyncio fails.

2. Localhost measurements mask real dynamics. Fallback: token-bucket rate limiting + artificial latency; validate by removing limiter and confirming results change.

7. WEEKLY PLAN

Week	Target	Owner
1 (Sep 1–7)	Metainfo format + file splitter; tracker skeleton (listen, accept, parse)	A, B
2 (Sep 8–14)	Tracker announce/peer-list complete; wire protocol messages + handshake	B, C
3 (Sep 15–21)	Single-peer E2E transfer (1 seeder → 1 leecher); piece hash verification	A, C
4 (Sep 22–28)	Piece selectors (×3); multi-peer integration (4 peers)	D, A+C
5 (Sep 29–Oct 5)	Scale to 6–8 peers; rate limiter; free-rider detection + choking	All
6 (Oct 6–12)	Experiment harness: automated runs, seed-controlled reproducibility	D, B
7 (Oct 13–19)	Run experiments (5 runs x 3 strategies x 4 configs); generate plots	D, All
8 (Oct 20–26)	Report, protocol spec, code cleanup, viva prep, hardening	All